

Andhra Pradesh – Grade 10 2016 – Mathematics

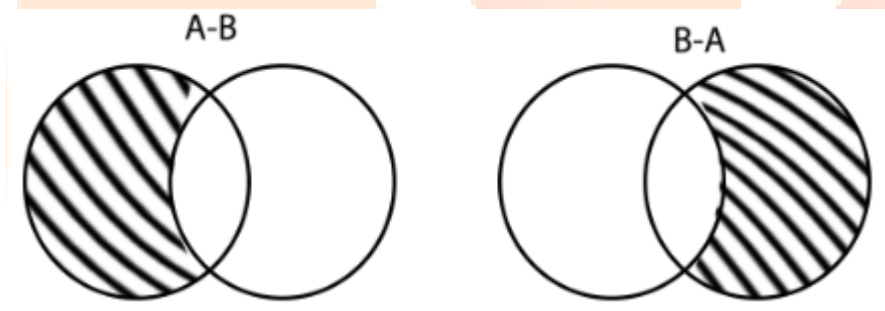
1. Expand $\log \frac{343}{125}$.

Solution:

$$\begin{aligned}\log \frac{343}{125} \\&= \log \frac{7^3}{5^3} \\&= \frac{\log 7}{\log 5} \\&= \frac{\log 7}{\log 5}\end{aligned}$$

2. Draw the Venn diagrams of the sets $(A - B)$, $(B - A)$.

Solution:



3. Find a quadratic polynomial, if the zeroes of it are 2 and -1 respectively.

Solution:

$$\text{Quadratic} = x^2 - (\text{sum})x + \text{product} = 0$$

$$\text{Sum} = 2 + (-1) = 1$$

$$\text{Product} = 2 \times -1 = -2$$

$$x^2 - x - 2 = 0$$

4. Find the roots of equations $2x^2 + x - 6 = 0$ by factorisation

Solution:

$$2x^2 + x - 6 = 0$$

$$2x^2 + 4x - 3x - 6 = 0$$

$$2x(x + 2) - 3(x + 2) = 0$$

$$x = -2 \quad \text{or} \quad x = \frac{3}{2}$$

5. 10 students of class X took part in a mathematics quiz. If the number of girls is four more than the number of boys; then find the number of boys and the number of girls, who took part in the quiz.

Solution:

Let number of boys be x

No. of girls be $10 - x$

No. of girls = 4 + No. of boys.

$$10 - x = 4 + x$$

$$6 = 2x$$

$$x = 3$$

No. of boys = 3

No. of girls = 7

6. Find the number of terms in the following AP. 7, 13, 19,, 205

Solution:

7, 13, 19,, 205

$$a_0 = 7$$

$$a_n = 205$$

$$d = 6$$

$$a_n = a_0 + (n-1)d$$

$$205 = 7 + (n-1)6$$

$$\frac{198}{6} = n - 1$$

$$33 + 1 = n$$

$$n = 34$$

7. Find the coordinates of the point, which divides the join of $(-1, 7)$ and $(4, -3)$ in the ratio $2 : 3$.

Solution:

$$m : n = 2 : 3$$

$$\begin{aligned} x, y &= \left\{ \frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n} \right\} \\ &= \frac{2 \times 4 + 3(-1)}{2+3}, \frac{2 \times -3 + 7 \times 3}{2+3} \\ &= \frac{5}{5}, \frac{-6+21}{5} \\ &= 1, 3 \end{aligned}$$

8. Find the area of the triangle, whose vertices are $(2, 0)$, $(1, 2)$, $(-1, 6)$. What do you observe?

Solution:

$$\begin{aligned} A &= \frac{1}{2} \{x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)\} \\ &= \frac{1}{2} \{2(2 - 6) + 1(6 - 0) + -1(0 - 2)\} \\ &= \frac{1}{2} \{-8 + 6 + 2\} = 0 \end{aligned}$$

We observe that since, Area = 0 hence, the point are collinear.

9. Find the value of $\log_{81} 3$.

Solution:

$$\begin{aligned} \log_{81} 3 \\ &= \frac{\log 3}{\log 81} \\ &= \frac{\log 3}{4 \log 3} = \frac{1}{4} \end{aligned}$$

10. List all the subsets of the following set $B = \{p, q\}$.

Solution:

All subsets of B are $\{\}, \{p\}, \{q\}, \{p, q\}$

11. Write the following set $\{x : x = 2n + 1 \text{ and } n \in \mathbb{N}\}$ in roster form.

Solution:

$\{3, 5, 7, 9, 11, 13, \dots\}$

12. If $p(x) = x^2 - 5x - 6$; find the value of $p(3)$.

Solution:

$$\begin{aligned} p(x) &= x^2 - 5x - 6 \\ p(3) &= 3^2 - 5 \times 3 - 6 \\ &= 9 - 15 - 6 \\ &= -12 \end{aligned}$$

13. Find the common ratio of GP $2, 2\sqrt{2}, 4, \dots$

Solution:

$$\begin{aligned} r &= \frac{2\sqrt{2}}{2} \\ r &= \sqrt{2} \\ \text{so, common ratio} &= \sqrt{2} \end{aligned}$$

14. Find the midpoint of the line segment joining the points (2, 7) and (12, - 7).

Solution:

$$= \frac{12+2}{2}, \frac{7-7}{2}$$

$$= (7, 0)$$

15. Show that $5-\sqrt{3}$ is irrational.

Solution:

Let $5 - \sqrt{3} = \frac{a}{b}$ be rational

$$a = 5b - \sqrt{3}b$$

$$\sqrt{3}b = 5b - a$$

On L.H.S $\sqrt{3}b$ is irrational

On R.H.S $5b - a$ is rational

i. e the assumption failed. Hence, $5 - \sqrt{3}$ is irrational.

16. If $A = \{1, 2, 3, 4\}$, $B = \{1, 2, 3, 5, 6\}$, then find (i) $A \cap B$, (ii) $B \cap A$ (iii) $A - B$, (iv) $B - A$, and what do you observe?

Solution:

$$(i) A \cap B \rightarrow \{1, 2, 3\}$$

$$(ii) B \cap A \rightarrow \{1, 2, 3\}$$

$$(iii) A - B \rightarrow \{4\}$$

$$(iv) B - A \rightarrow \{5, 6\}$$

17. Find the zeroes of the polynomial $p(x) = x^2 - 4x + 3$ and verify the relationship between zeroes and coefficients.

Solution:

$$x^2 - 4x + 3 = 0$$

$$x^2 - 3x - x + 3 = 0$$

$$x(x-3) - 1(x-3) = 0$$

$$x = 3, 1$$

$$\text{sum of zeroes} = 3 + 1 = 4$$

$$\rightarrow \frac{-b}{a} = -\left(\frac{-4}{1}\right) = 4$$

$$\text{product} \rightarrow 3 \times 1 = 3$$

$$\rightarrow \frac{c}{a} = \frac{3}{1} = 3$$

18. Solve the quadratic equation $2x^2 - x + 4 = 0$ by completing the square.

Solution:

$$2x^2 + x - 4 = 0$$

$$x^2 + \frac{x}{2} - 2 = 0$$

$$x^2 + \frac{x}{2} + \left(\frac{1}{4}\right)^2 = 2 + \left(\frac{1}{4}\right)^2$$

$$\left(x + \frac{1}{4}\right)^2 = \frac{33}{16}$$

$$x + \frac{1}{4} = \frac{\sqrt{33}}{4}$$

$$x = \frac{\sqrt{33}}{4} - \frac{1}{4}$$

$$= \frac{\sqrt{33} - 1}{4}$$

19. Solve the equations.

$$\frac{10}{x+y} + \frac{2}{x-y} = 4, \frac{15}{x+y} - \frac{5}{x-y} = -2$$

Solution:

let

$$\frac{1}{x+y} = k$$

$$\frac{1}{x-y} = m$$

$$\rightarrow 10k + 2m = 4 \times 5$$

$$15k - 5m = -2 \times 2$$

$$\rightarrow 50k + 10m = 20$$

$$30k - 10m = -4$$

$$\begin{array}{r} 50k + 10m = 20 \\ 30k - 10m = -4 \\ \hline 80k = 16 \end{array}$$

$$k = \frac{16}{80} = \frac{1}{5}$$

$$\frac{1}{x+y} = \frac{1}{5} \Rightarrow x+y=5 \dots\dots\dots(1)$$

$$m = \frac{4-10k}{2} = \frac{4-2}{2} = 1$$

$$\frac{1}{x+y} = \frac{1}{1} \Rightarrow x-y=1 \dots\dots\dots(2)$$

$$x+y=5$$

$$\underline{x-y=1}$$

$$2x=6$$

$$x=3$$

$$y = 5 - x$$

$$= 5 - 3$$

$$y = 2$$

20. Solve the pair of equations by Elimination method.

$$2x + y - 5 = 0, 3x - 2y - 4 = 0$$

Solution:

$$2x + y = 5 \times 2$$

$$3x - 2y = 4$$

$$\rightarrow 4x + 2y = 10$$

$$\underline{3x - 2y = 4} \quad (\text{on adding})$$

$$7x = 14$$

$$x = 2$$

$$y = 5 - 2x$$

$$= 5 - 2 \times 2$$

$$= 5 - 4 = 1$$

21. If the sum of the first 7 terms of an AP is 49 and that of 17 terms is 289; find the sum of the first n terms.

Solution:

$$\text{Sum} = \frac{n}{2} [2a_0 + (n-1)d]$$

$$49 = \frac{7}{2} [2a_0 + (7-1)d]$$

$$14 = 2a_0 + 6d$$

$$7 = a_0 + 3d \quad \dots(1)$$

$$289 = \frac{17}{2} [2a_0 + (17-1)d]$$

$$34 = 2a_0 + 16d$$

$$17 = a_0 + 8d \quad \dots(2)$$

$$7 = a_0 + 3d$$

$$17 = a_0 + 8d$$

$$\begin{array}{r} - \quad - \quad - \\ \hline \end{array}$$

$$10 = 5d$$

$$d = 2$$

$$a_0 = 7 - 3d$$

$$= 7 - 3 \times 2 = 1$$

Sum of first n terms

$$= \frac{n}{2} [2a_0 + (n-1)a]$$

$$= \frac{n}{2} [2 \times 1 + (n-1)2]$$

$$= \frac{n}{2} [2 + 2n - 2]$$

$$= \frac{n}{2} \times 2n = n^2$$

22. Find the area of triangle formed by joining the mid points of the sides of the triangle, whose vertices are (0, -1); (2, 1) and (0, 3). Find the ratio of this area to the area of the given triangle.

Solution:

midpoints

$$D = \frac{0+2}{2}, \frac{(-1+1)}{2}$$

$$= (1, 0)$$

$$E = \frac{2+0}{2}, \frac{(1+3)}{2}$$

$$= (1, 2)$$

$$F = \frac{0+0}{2}, \frac{(3-1)}{2}$$

$$= (0,1)$$

$$\text{Area of } \Delta = \frac{1}{2} \{ 0(1-3) + 2(3+1) + 0(-1-1) \}$$

$$= \frac{1}{2} \{ 8 \} = 4$$

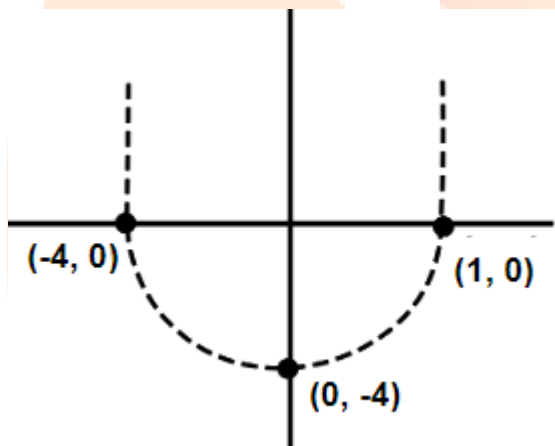
$$\text{Area of midpoint } \Delta = \frac{1}{2} \{ 1(2-1) + 1(1-0) + 0(0-2) \}$$

$$= \frac{1}{2} \{ 1 \} = 1$$

$$\frac{\text{Area of midpoint}}{\text{Area of } \Delta} = \frac{1}{4} \Rightarrow 1:4$$

23. Draw the graph of $p(x) = x^2 + 3x - 4$ and find zeroes. Verify the zeroes of the polynomials.

Solution:



@ y intercept $x = 0$

$$y = 0 + 0 - 4 = -4 \quad (0, -4)$$

@ x intercept $y = 0$

$$0 = x^2 + 3x - 4$$

$$= x^2 + 4x - x - 4$$

$$= x(x+4) - 1(x+4)$$

$$x = -4, x = 1 \quad (-4, 0) (1, 0)$$

24. Solve the following equations graphically.

$$3x - y = 7,$$

$$2x + 3y = 1$$

Solution:

$$y = 3x - 7$$

x	0	1	2	3
y	-7	-4	-1	2

$$y = \frac{1-2x}{3}$$

x	-1	2	-4	0
y	1	-1	3	$\frac{1}{3}$

